Joan Ribot Oliver

Exercise 2: PDF report and documentation

Abstract

This report presents a statistical study of nights lost to bad weather for several years in a certain abstronomical observatory. There are two datasets provided for analysis: one containing yearly percentages of lost nights, to which a constant and a linear trend model are fitted, and another with monthly averages, to which a constant and a sinusoidal model with an annual period are fitted with different confidence intervals for each case.

All model fittings are performed using the chi-squared minimization method in R [1]. Constant models are found to be inconsistent with the data in both cases, while linear and sinusoidal models provide significantly better fits for the respective datasets according to the goodness of fit (Q) values calculated and also to the F-test statistic in the first case.

1 Introduction

The two datasets analysed in this report were provided by the professor Maite Ceballos the class of the 29th of November and 1st of December of 2025 as part of the course assignment. All data has ben read from the provided .dat files without any further transformations.

The percentages of nights lost to bad weather for a number of years in an astronomical observatory are given in the file `Exercise2_Data_percentage_years.dat`. A constant model ($y = a$) is first considered to describe the data with a confidence interval at 90% using the weighted (inverse of the variance) chi-squared variation ($\Delta\chi^2 = 2.71$ for 1 parameter, the constant a) method [2]. The goodness of the fit is calculated as the probability that χ^2 is greater than the measured value under the assumption that the model is correct.

Afterwards, a simple climate change model is considered and a linear trend is assumed

$$y = a + bt. \tag{1}$$

The best-fit values of the two parameters of the linear model are estimated minimizing χ^2, and again the confidence intervals at 90% ($\Delta\chi^2 = 2.71$ for one parameter) and 99% ($\Delta\chi^2 = 6.63$) confidence for the slope, b, of this fit are provided.

Finally for the first dataset, the improvement in χ^2 introduced by the linear model with respect to the constant model is evaluated using the F-test statistic as seen in class, Which determines that the improvement of the linear model over the constant model is statistically significant.

The second dataset considers monthly precentage averages of nights lost to bad weather in the file `Exercise2_Data_percentage_months.dat`. To demonstrate that the data are not compatible with no seasonal variation (month to month), a constant model is fitted and the goodness of fit is evaluated.

A new sinusoidal model with an annual period (12 months) and a constant term is then considered to describe the seasonal variation of the data:

$$f = f_c + f_e \sin\left(2\pi\frac{(t-t_0)}{\tau}\right) \tag{2}$$

where $\tau = 12$ months, which is the period of the sinusoid.

The best-fit values of the three parameters of this model (the constant f_c, the amplitude of the sinusoidal function f_e and the origin t_0) are calculated minimizing χ^2.

The value of the amplitude of the seasonal component, f_e, is estimated along with its confidence interval at 90% ($\Delta\chi^2 = 2.71$) confidence independently of the other two parameters.

Finally, considering the two amplitude parameters f_c and f_e simultaneously, the area in that parameter space that contains 95.4% confidence ($\Delta\chi^2 = 6.17$, 2σ for two parameters) is delimited using the method of χ^2 variation [2].

2 Results

2.1 Yearly variation: percentage of nights lost to bad weather per year

The yearly variation of the percentage of nights lost to bad weather dataset provides 22 data points, from 1984 to 2005. For this analysis, two models are considered: a constant model and a linear trend model.

The goodness of fit for each model is evaluated by calculating the p-value, which is the probability that the χ^2 value is larger than the measured value if the model were correct. The $\Delta\chi^2$ method is used to estimate confidence intervals for the model parameters.

2.1.1 Constant model

The constant model fitted to the data of the percentage of nights lost to bad weather per year can be seen in Figure 1.

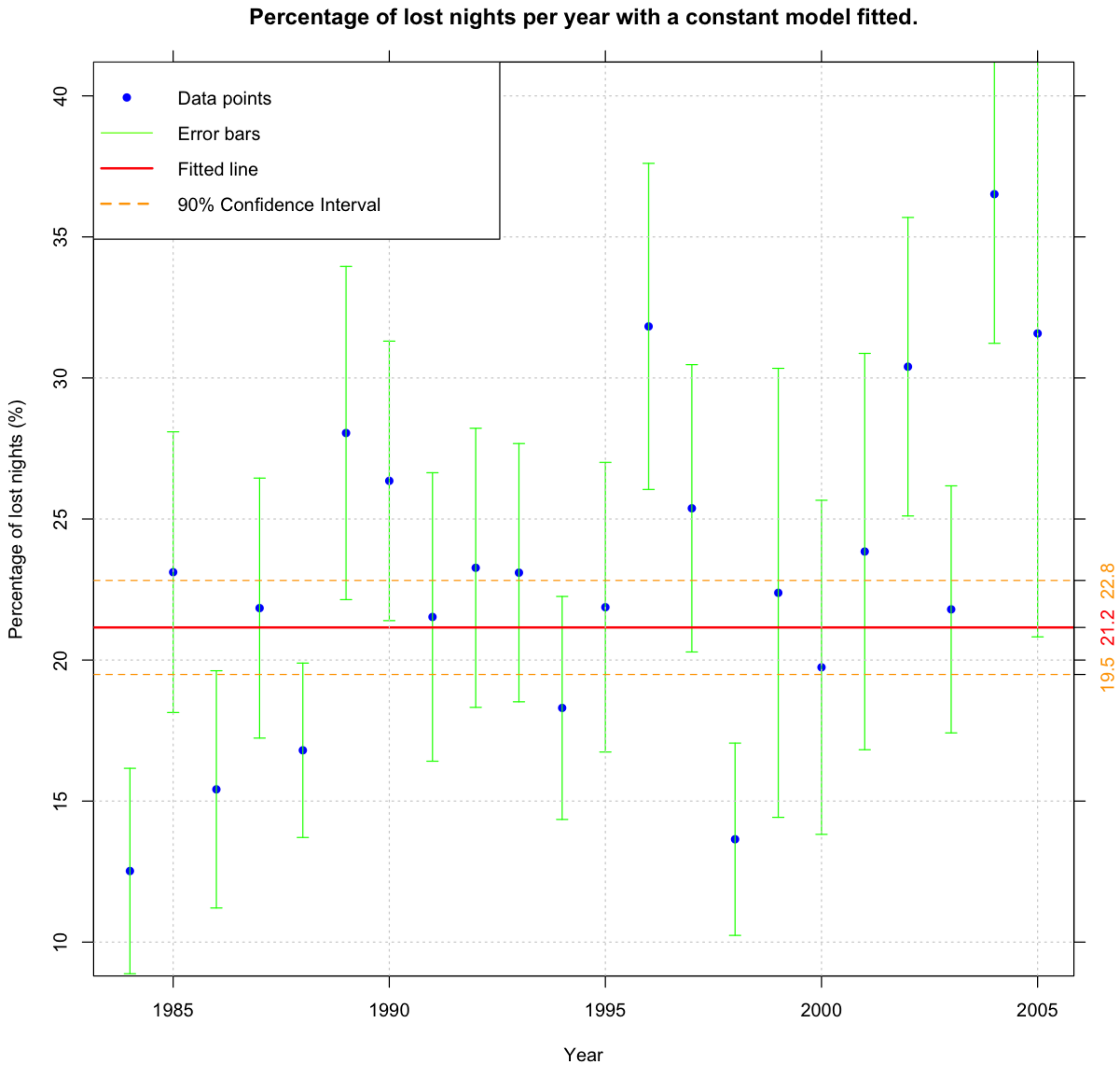

Figure 1: Constant model fitted to the yearly percentage of nights lost to bad weather data. Blue points represent the data (with error bars) and the red line indicates the best-fit constant value.

The best-fit value of the constant is obtained by minimizing the χ^2. In this case, the error on the constant parameter is estimated using the $\Delta\chi^2$ method at 90% confidence level.

a	χ^2_{min}	ν (dof)	$\chi^2_{reduced}$	Q
21.2 ± 1.7	34.652	21	1.650	0.031

Table 1: Constant, minimum chi-squared, degrees of freedom, reduced chi-squared and goodness of fit for the constant model fitted to the yearly percentage of nights lost to bad weather data.

The low value of the goodness of fit ($Q < 0.05$) indicates that the constant model is not a good fit for the data, according to the discrepancies in Figure 1 between the data points and the fitted constant line.

2.1.2 Linear model

Then a linear trend model is fitted to the same data in order to account for possible climate change effects with the shape of Equation 1. The best-fit values of the two parameters (slope, b, and intercept, a) are obtained again by minimizing the χ^2 statistic and the goodness of fit is evaluated similarly as before. Figure 2 shows the linear model fitted to the data along with the confidence intervals for the slope parameter.

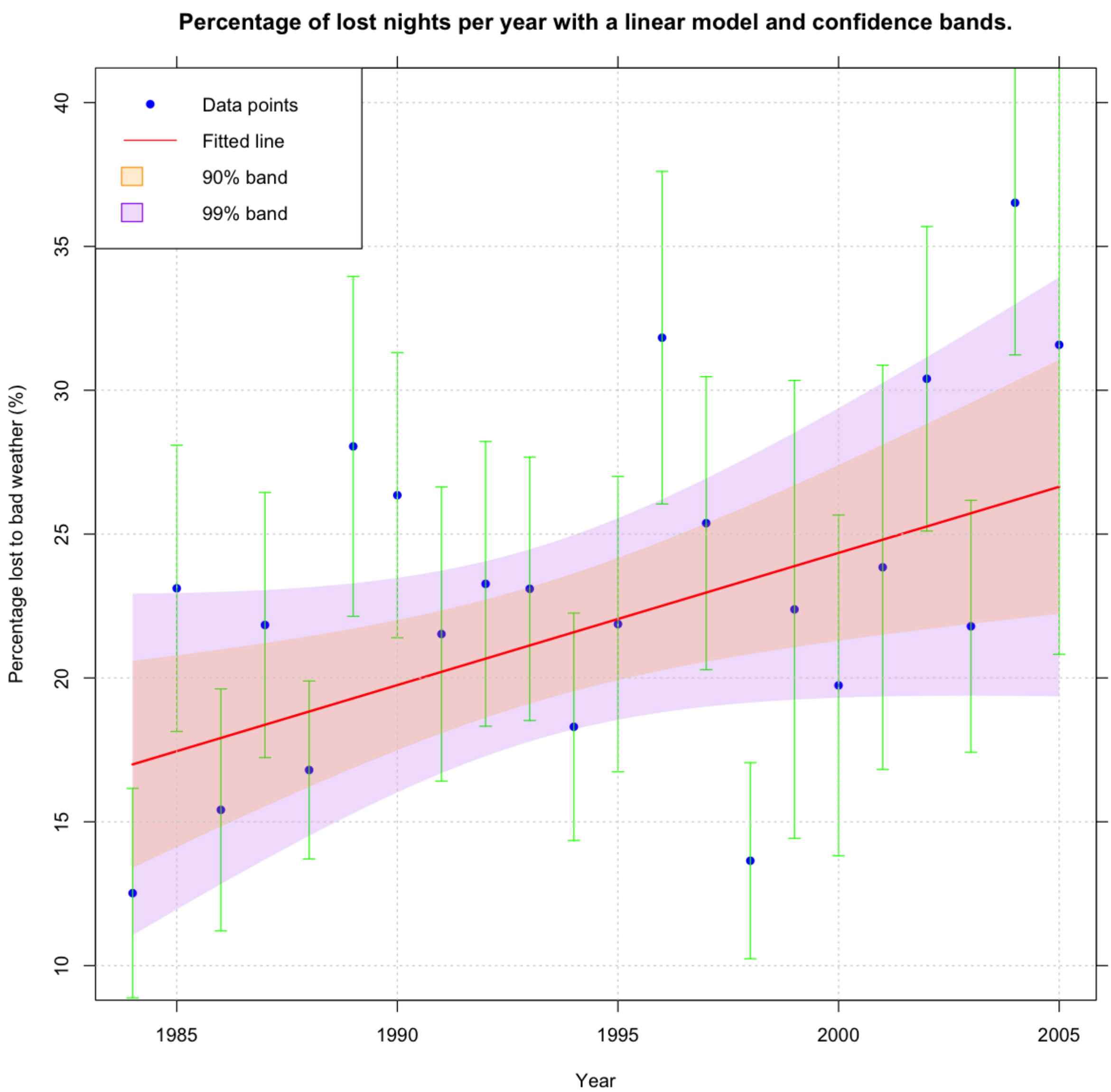

Figure 2: Linear model fitted to the yearly percentage of nights lost to bad weather data. The blue points represent the data with error bars, while the red line indicates the best-fit linear trend. The shaded areas represent the 90% and 99% confidence intervals for the slope parameter.

The procedure followed to calculate the confidence intervals for the slope parameter is the one seen in class.

The summary of the chi-squared statistics for the linear model is presented in Table 2.

a	b	χ^2_{min}	ν (dof)	$\chi^2_{reduced}$	Q
-894	0.460	26.843	20	1.342	0.140

Table 2: Constant, slope, minimum chi-squared, degrees of freedom, reduced chi-squared and goodness of fit for the linear model fitted to the yearly percentage of nights lost to bad weather data.

The errors on the slope parameter are estimated using the $\Delta\chi^2$ method at 90% and 99% confidence levels, resulting in:

$$b_{90\%} = 0.460 \pm 0.329 \implies b_{90\%}\epsilon[0.131, 0.789],$$
$$b_{99\%} = 0.460 \pm 0.542 \implies b_{99\%}\epsilon[-0.082, 1.002].$$

The higher goodness of fit value ($Q = 0.140 > 0.05$) for this case indicates that the linear model provides a better fit for the data than the constant model.

2.1.3 Model comparison

The improvement in χ^2 introduced by the linear model with respect to the constant model is evaluated using the F-test statistic, also seen in class.

Model	χ^2_{min}	ν (dof)	$\chi^2_{reduced}$	Q	F	p
Constant	34.652	21	1.650	0.031	5.819	0.026
Linear	26.843	20	1.342	0.140		

Table 3: Summary of chi-squared statistics for the constant and linear models fitted to the yearly percentage of nights lost to bad weather data, including the F-test statistic and p-value for model comparison.

The F-test statistic value shows a significant improvement of the linear model over the constant model, a reduction of 7.81 points in chi-squared. The small p-value (< 0.05) rejects the null hypothesis that both models fit the data equally well. Thus, the linear model is considered a better fit for the data than the constant model.

2.2 Seasonal variation: average percentage of nights lost to bad weather per month

The seasonal variation (month to month) of the percentage of nights lost to bad weather dataset provides 12 data points, one for each month. For this analysis, two models are considered: a constant model and a sinusoidal model with an annual period.

2.2.1 Constant model

First, a constant model is fitted to the monthly average data as shown in Figure 3.

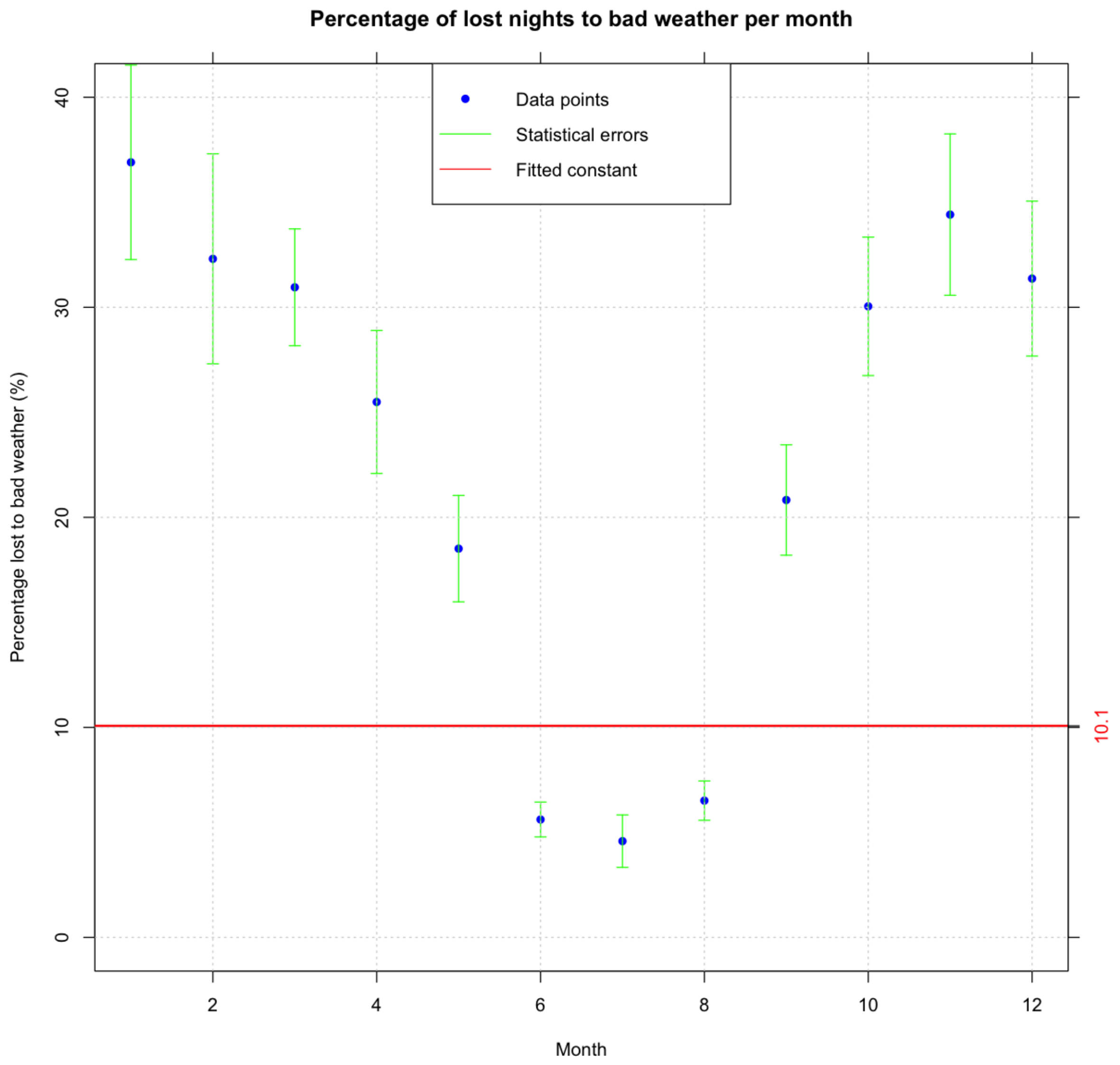

Figure 3: Constant model fitted to the monthly percentage of nights lost to bad weather data. Blue points represent the data with error bars, while the red line indicates the best-fit constant value.

The best-fit value of the constant is obtained by minimizing the χ^2 statistic. The goodness of fit is evaluated by calculating the probability that the χ^2 value is larger than the measured value if the model were correct.

a	χ^2_{min}	ν (dof)	$\chi^2_{reduced}$	Q
10±5	331.172	11	30.107	0

Table 4: Constant, minimum chi-squared, degrees of freedom, reduced chi-squared and goodness of fit for the constant model fitted to the monthly percentage of nights lost to bad weather data.

The very low value of the goodness of fit ($Q = 0$) indicates that the constant model is not a good fit for the data, as can be seen by the discrepancies in Figure 3 between the data points and the fitted constant line. Then, the null hypothesis (H0), which states that there is no seasonal variation, is rejected.

2.2.2 Sinusoidal model

Then a sinusoidal model with an annual period is fitted to the data following the formula in Equation 2. The initial guesses for the parameters are set according to some values given by Maite Ceballos after the exam, and are the following: $f_c = 18$, $f_e = 20$ and $t_0 = 10$.

The initial model is shown in Figure 4 and it can be seen that it is a poor fit.

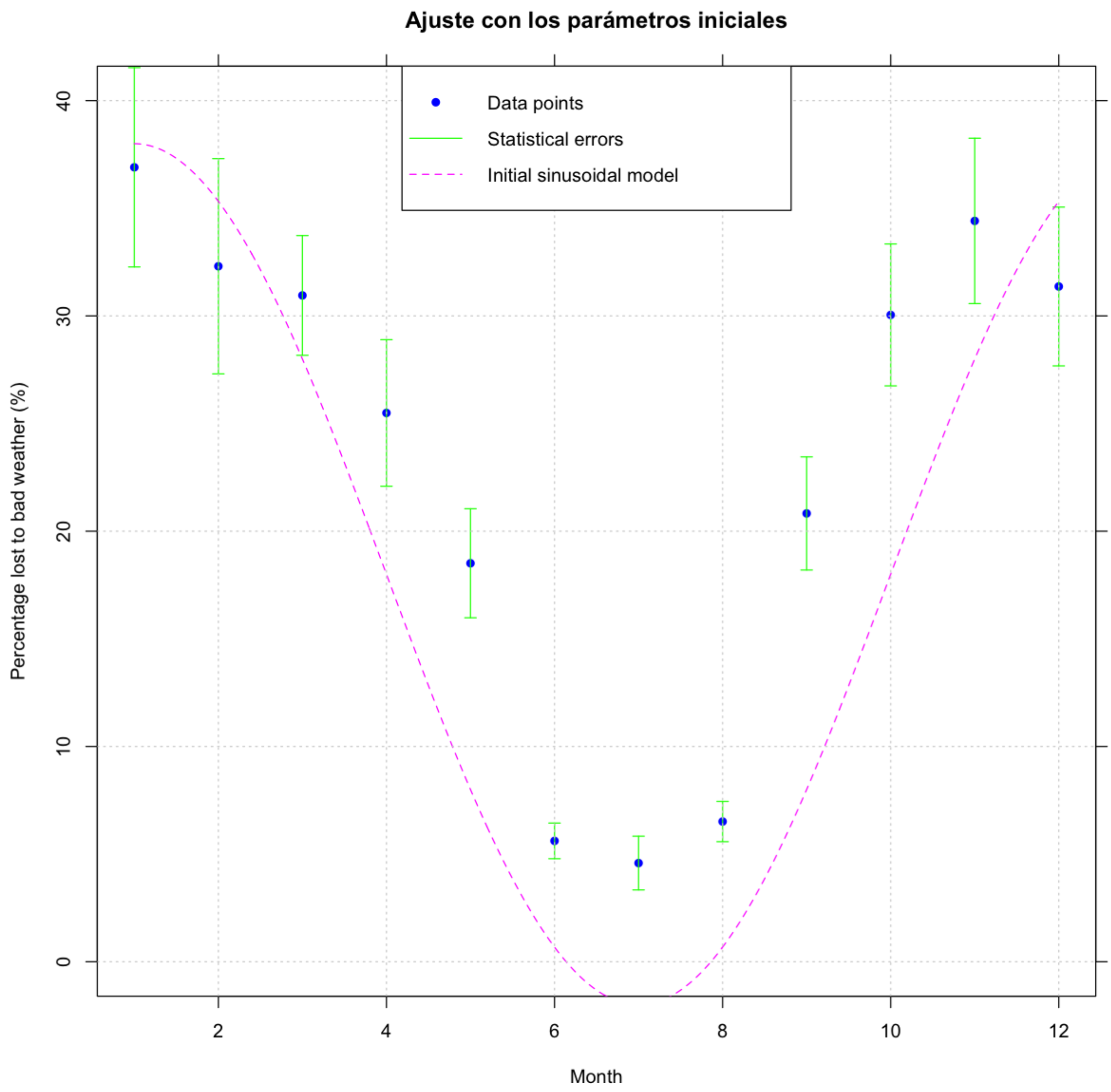

Figure 4: Sinusoidal model fitted to the monthly percentage of nights lost to bad weather data with initial parameter. Blue points represent the data with error bars, while the magenta line indicates the initial sinusoidal model.

The best-fit values of the three parameters (f_c, f_e, t_0) with their 90% confidence errors are obtained by minimizing the χ^2 statistic and summarized in Table 5.

f_c	f_e	t_0	χ^2_{min}	ν (dof)	$\chi^2_{reduced}$
22.6±1.4	18.2±1.7	9.87±0.16	24.369	11	2.215

Table 5: Best-fit parameters with 90% confidence errors, minimum chi-squared, degrees of freedom, reduced chi-squared for the sinusoidal model fitted to the monthly average percentage of nights lost to bad weather data.

Finally, it can be seen in Figure 5 how well the sinusoidal model fits the data.

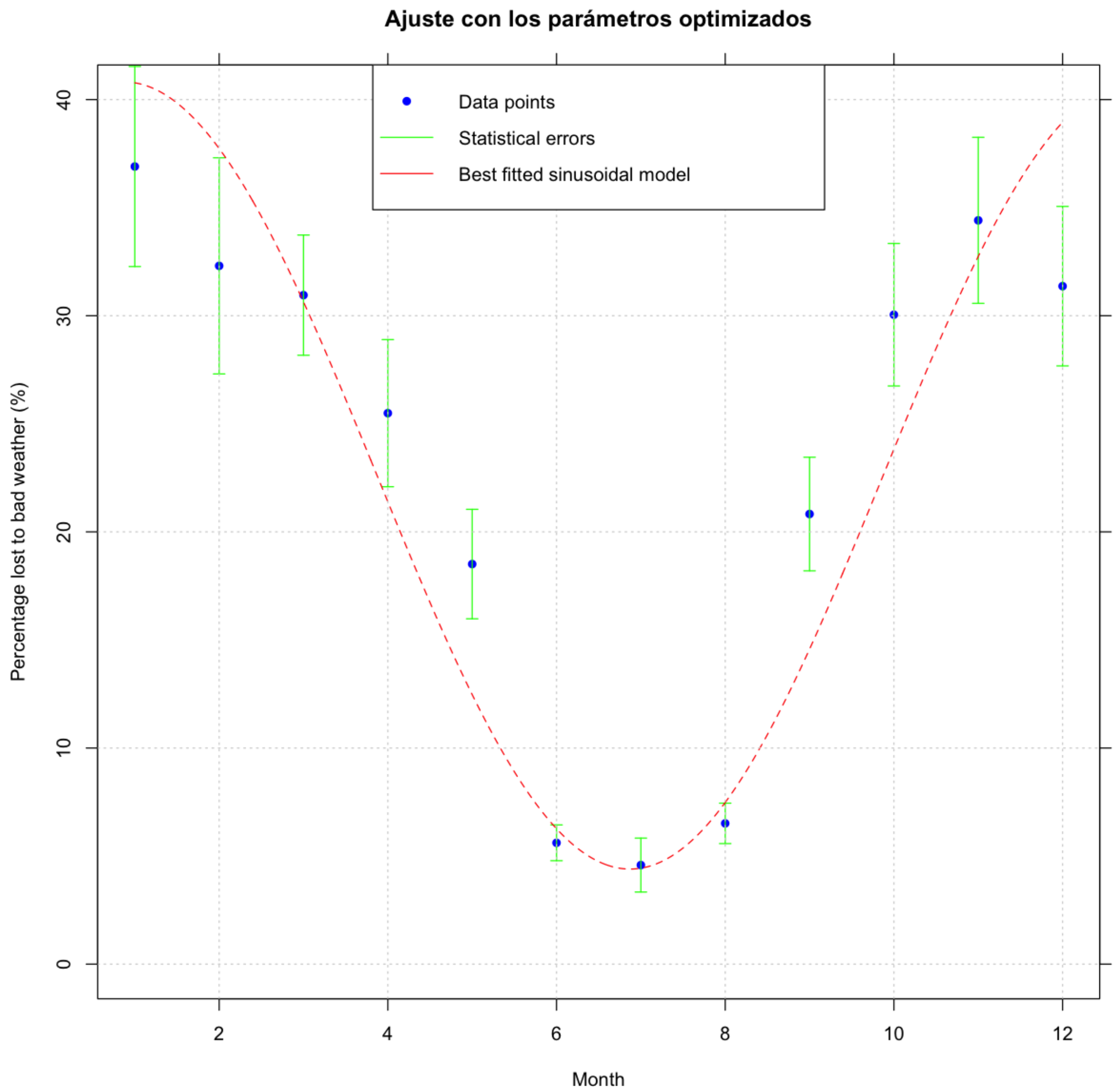

Figure 5: Sinusoidal model fitted to the monthly percentage of nights lost to bad weather data. The blue points represent the data with error bars, while the red line indicates the best-fit sinusoidal model.

2.2.3 Model parameters confidence intervals

Considering the two amplitude parameters f_c and f_e simultaneously, the area in that parameter space that contains 95.4% confidence is delimited using the method of χ^2 variation. Figure 6 shows the confidence region for the amplitude parameters f_c and f_e of the sinusoidal model fitted to the monthly average percentage of nights lost to bad weather data.

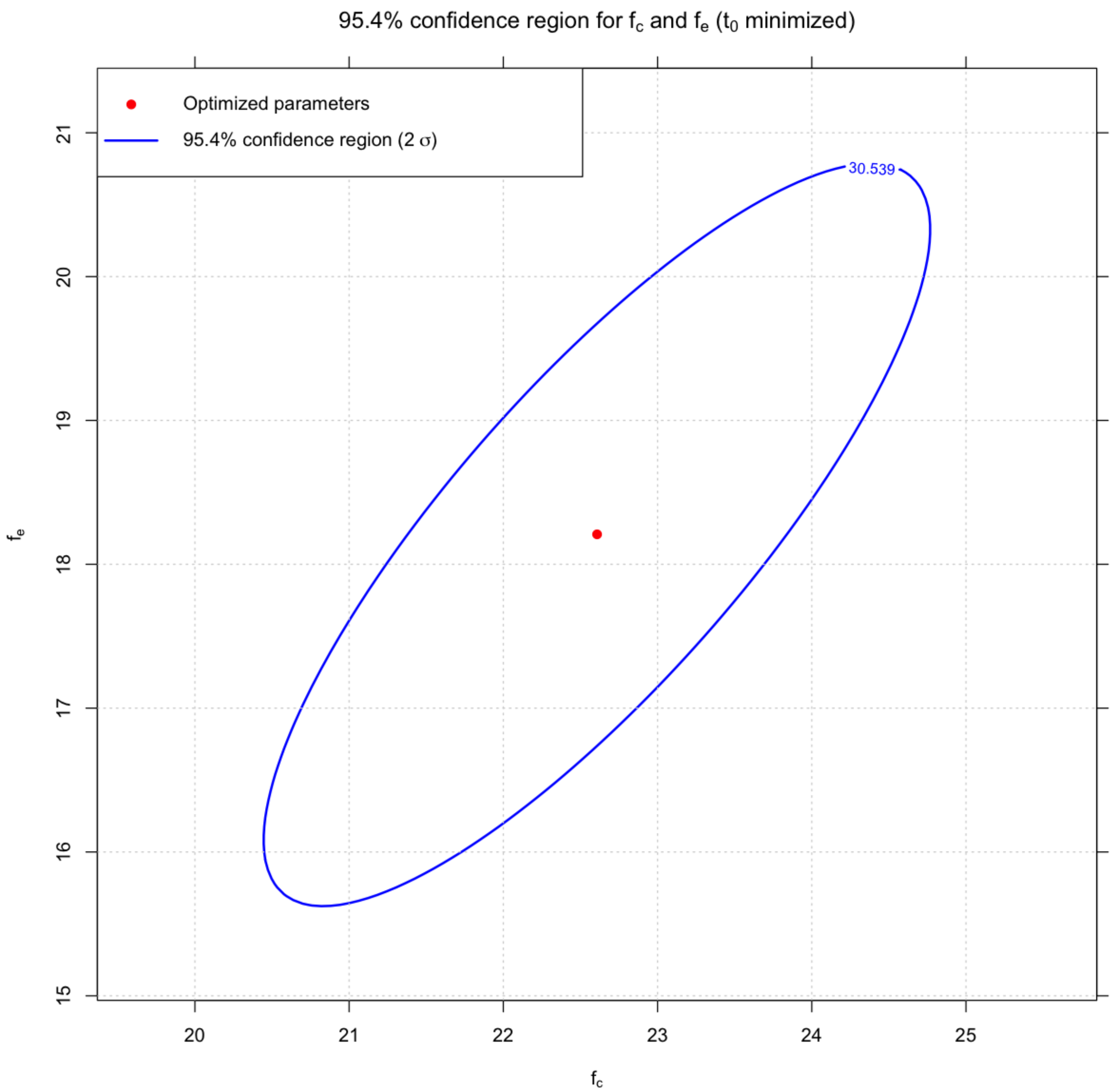

Figure 6: Confidence region for the amplitude parameters f_c and f_e of the sinusoidal model fitted to the monthly average percentage of nights lost to bad weather data. The red dot represents the best-fit values, while the blue contour represents the 95.4% (2σ) confidence region where $\Delta\chi^2 \leq 6.17$.

3 Conclusions

The analysis of yearly percentage of nights lost to bad weather is shown to have a significant linear trend, suggesting a yearly increase in the percentage of nights lost. The constant model is rejected based on its low goodness of fit value, and the F-test statistic confirms that the linear model provides a significantly better fit.

The seasonal variation analysis demonstrates a strong monthly variation in the percentage of nights lost to bad weather. The constant model is rejected due to its very low goodness of fit value, then it is not compatible with no seasonal variation. The sinusoidal model with an annual period fits the data very well and has a statistically significant seasonal component. The confidence intervals for the amplitude parameters are estimated and the 95.4% confidence region is delimited.

The chi squared minimization method proves to be an effective tool for fitting models to the data and estimating parameter uncertainties, and the goodness of fit evaluations provide valuable insights into the adequacy of the models used, among other statistical tests.

References

[1] R Core Team, *R: A Language and Environment for Statistical Computing*, R Foundation for Statistical Computing, Vienna, 2025. `https://www.R-project.org/`

[2] W. H. Press, S. A. Teukolsky, W. T. Vetterling and B. P. Flannery, *Numerical Recipes: The Art of Scientific Computing*, 3rd ed., Cambridge University Press, Cambridge, 2007. ISBN 978-0-521-88068-8.